J. Liebig, NOAA GLERL, 2001

## *Bythotrephes longimanus*

**Common name:** spiny waterflea

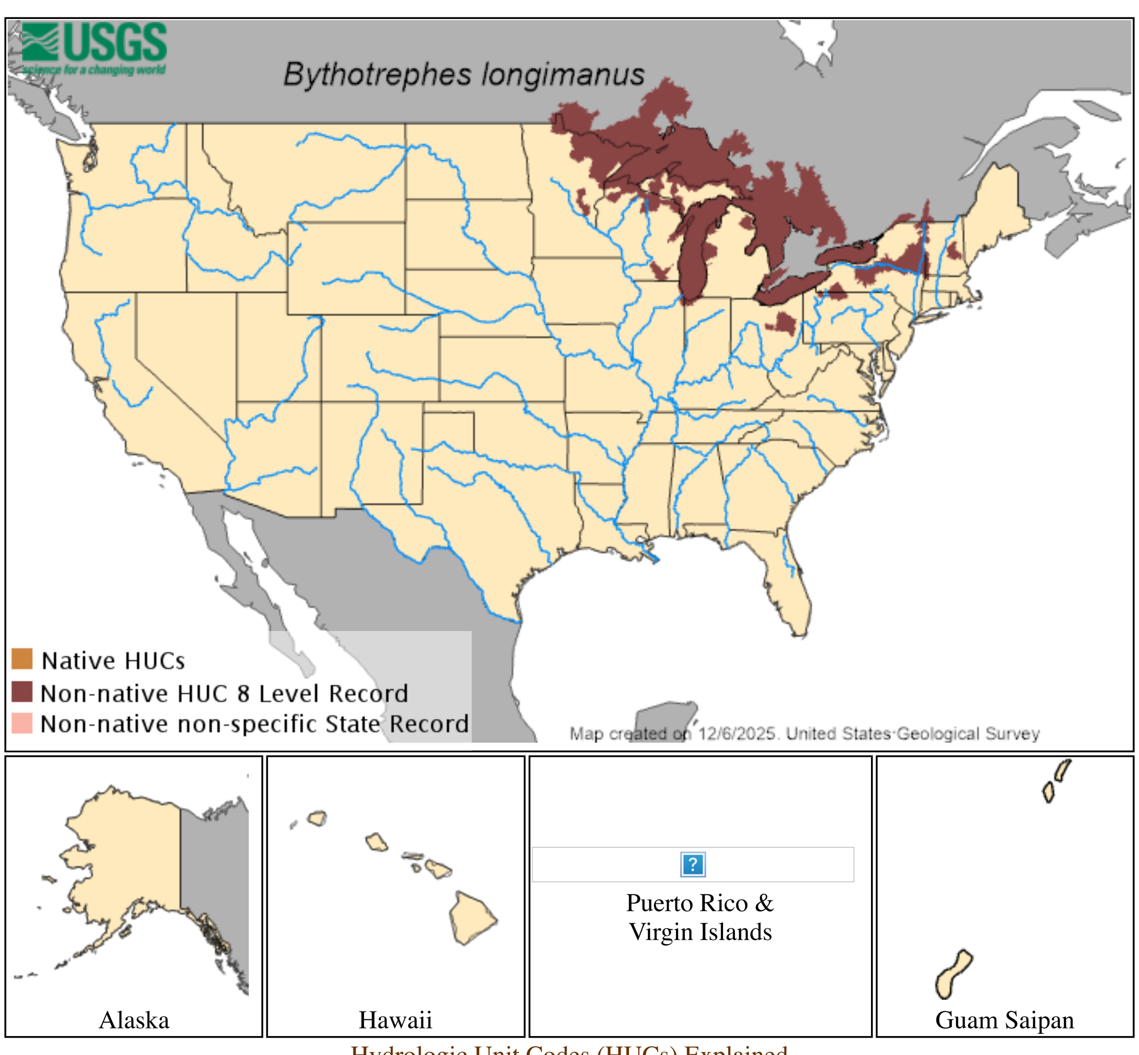
**Synonyms and Other Names:** Formerly known as *Bythotrephes cederstroemii* (Berg and Garton, 1994; Yan and Pawson, 1998; Berg et al., 2002; Therriault et al., 2002). spiny water-flea

**Taxonomy:** available through ITIS

**Identification:** *Bythotrephes longimanus* is a large cladoceran distinguished by a long straight tail spine that is twice as long as its body and has one to three pairs of barbs. Parthenogenically produced animals have kink in middle of their spine and sexually produced animals lack the kink. *Bythotrephes* appearance is similar to *Cercopagis pengoi*, another Great Lakes invader, except *Bythotrephes* is larger with a more robust spine that lacks a hook at the end.

**Size:** can reach 15 mm

**Native Range:** Northern Europe and Asia




Hydrologic Unit Codes (HUCs) Explained
Interactive maps: Point Distribution Maps

**Nonindigenous Occurrences**:

Table 1. States with nonindigenous occurrences, the earliest and latest observations in each state, and the tally and names of HUCs with observations†. Names and dates are hyperlinked to their relevant specimen records. The list of references for all nonindigenous occurrences of *Bythotrephes longimanus* are found here.

| State | First Observed | Last Observed | Total HUCs with observations† | HUCs with observations† |
|---|---|---|---|---|
| IL | 1986 | 2011 | 3 | Chicago; Lake Michigan; Little Calumet-Galien |
| IN | 1986 | 1988 | 1 | Lake Michigan |
| MI | 1984 | 2020 | 12 | Betsie-Platte; Betsy-Chocolay; Dead-Kelsey; Keweenaw Peninsula; Lake Huron; Lake Michigan; Lake Superior; Michigamme; Ontonagon; Pere Marquette-White; St. Marys; Thunder Bay |
| MN | 1987 | 2022 | 12 | Baptism-Brule; Big Fork; Cloquet; Lake of the Woods; Lake Superior; Little Fork; Lower Rainy; Rainy Headwaters; Rainy Lake; Rum; St. Louis; Vermilion |
| NH | 2023 | 2025 | 2 | Pemigewasset River; Winnipesaukee River |
| NY | 1985 | 2022 | 11 | Hudson-Hoosic; Lake Champlain; Lake Erie; Lake Ontario; Mettawee River; Mohawk; Oneida; Sacandaga; Seneca; Upper Allegheny; Upper Hudson |
| OH | 1985 | 2019 | 3 | Lake Erie; Tuscarawas; Walhonding |
| PA | 1998 | 2013 | 2 | Lake Erie; Upper Allegheny |
| VT | 2014 | 2025 | 1 | Lake Champlain |
| WI | 1986 | 2019 | 12 | Bad-Montreal; Beartrap-Nemadji; Door-Kewaunee; Flambeau; Lake Michigan; Lake Superior; Lower Fox; Menominee; Middle Rock; Namekagon; St. Louis; Upper Wisconsin |

Table last updated 12/8/2025

† Populations may not be currently present.

**Ecology:** *Bythotrephes longimanus* is found among the zooplankton in the upper water column of large and small temperate lakes. They can tolerate brackish water, and are most abundant in late summer and autumn. Occurrence and density of *Bythotrephes* populations are determined mainly by water temperature and salinity; *Bythotrephes* is limited to regions where water temperature ranges between 4 and 30°C and salinity is between 0.04 and 8.0 parts per thousand, but it prefers temperature between 10 and 24°C and salinity between 0.04 and 0.4 ppt (Grigorovich et al. 1998). Temperature plays a major role in determining the abundance and location of *Bythotrephes* in the Great Lakes, as they prefer cooler water and cannot tolerate very warm lake temperatures (Berg and Garton 1988, Garton et al. 1990, Brown and Branstrator 2004). *Bythotrephes* occurs in oligotrophic and mesotrophic lakes and has a lower tolerance to low dissolved oxygen concentrations than the native cladoceran *Leptodora kindtii* (Sorensen and Branstrator, 2017). *Bythotrephes* can reproduce both asexually and sexually; unfertilized eggs are carried in a brood pouch, and fertilized eggs are cast in the fall, hatching the following spring (Evans, 1988). The intensity and type of predation pressure appears to affect the size of *Bythotrephes*, its spine length, and the extent of its diel migrations (Straile and Halbich, 2000).

*Bythotrephes longimanus* is a visual predator, using its large compound eye to detect zooplankton (Azan et al., 2015). They consume 75% of their body weight each day in prey items (Lehman et al., 1997).

**Means of Introduction:** *Bythotrephes* was probably introduced from ship ballast water (Sprules et al. 1990, Berg et al. 2002) and possibly as diapausing eggs from sediment in ballast tanks (Evans 1988).

**Status:** *Bythotrephes* is established in all of the Great Lakes and many inland lakes in the region. Densities are very low in Lake Ontario, low in southern Lake Michigan and offshore areas of Lake Superior, moderate to high in Lake Huron, and very high in the central basin of Lake Erie (Barbiero et al., 2001; Vanderploeg et al., 2002; Brown and Branstrator, 2004).

**Impact of Introduction:**

**Summary of species impacts derived from literature review. Click on an icon to find out more...**

| Ecological | Economic | Human Health | Other |
|---|---|---|---|

It has caused major changes in the zooplankton community structure; invasion history; reproduce rapidly; competes directly with small fish and can have impact on zooplankton community (USEPA 2008).

The first noticeable impact of *Bythotrephes* was to fisherman. The tail spines of *Bythotrephes* hook on fishing lines, fouling fishing gear. *Bythotrephes* consumes small zooplankton such as small cladocerans, copepods, and rotifers, competing directly with planktivorous larval fish for food (Berg and Garton 1988, Evans 1988, Vanderploeg et al. 1993). There is concern that Bythotrephes could increase bioaccumulation of contaminants by lengthening the food chain (Vanderploeg et al. 2002). *Bythotrephes* has been implicated as a factor in the decline of alewife (*Alosa pseudoharengus*) in Lakes Ontario, Erie, Huron, and Michigan (Evans 1988). *Bythotrephes* also competes with, and possibly preys on, *Leptodora kindtii* and may be a causal factor in the decline of *Leptodora* (Branstrator 1995). *Bythotrephes* and *Leptodora* abundances are often negatively correlated (Garton et al. 1990, Branstrator 1995). There is speculation that *Bythotrephes* may control the abundance of *Cercopagis pengoi* though competition and predation (Vanderploeg et al. 2002). *Bythotrephes* is a food source for fish including yellow perch, white perch, walleye, white bass, alewife, bloater chub, chinook salmon, emerald shiner, spottail shiner, rainbow smelt, lake herring, lake whitefish and deepwater sculpin (Bur et al. 1986, Makarewicz and Jones 1990, Branstrator and Lehman 1996).

**Remarks:** Deweese et al (2021) report on preliminary microfossil evidence for the presence of Bythotrephes in 4 inland lakes of the Great Lakes region which would place the introduction dates for this species signficantly earlier. The authors still speculate whether the data may be an artefact (albeit of a common method), and should be considered preliminary at best. If these corroborated, earliest dates of reported microfossils should be 1934 (Lake Kabetogama, MN), 1908 (Mille Lacs, MN), 1900 (Lake Nipissing, Ontario) and 1650s (Three Mile Lake, Ontario). Such early records call into question both the presumed means of introduction for this species (ballast water) as well as the possibility that the species should be considered native.

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**Other Resources:**
http://www.seagrant.umn.edu/exotics/spiny.html

*Bythotrephes longimanus* [spiny water flea] (ANS Clearinghouse Bibliography)

*Bythotrephes longimanus* (Global Invasive Species Database)

NY Invasive Species Clearinghouse

GLIFWC-Maps

Great Lakes Water Life

US Fish and Wildlife Service Ecological Risk Screening Summary for *Bythotrephes longimanus*

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